The Job Sequencing and Tool Switching Problem

Part III: SSP–GTSP Structural Equivalences

Scope. This document establishes the formal structural relationship between the U-SSP and the Generalized Traveling Salesman Problem (GTSP). We prove: (1) a forward polynomial reduction from SSP to GTSP via the Intersecting-to-Nonintersecting (I-N) transformation; (2) a *Disjointness Condition* identifying when SSP is natively a GTSP; (3) a *Metric Barrier* showing that the reverse reduction does not hold in general. These results justify using GTSP solvers (Concorde, LKH) on the configuration graph.

Contents

The GTSP and the Overlap Problem

Definition 1.1 (Generalized TSP (GTSP))**.** Given a complete graph on vertex set V partitioned into disjoint clusters $V_1, \ldots, V_n$ with edge costs $w : V \times V \to \mathbb{R}_{\geq 0}$, find a minimum-cost Hamiltonian cycle that visits *exactly one vertex from each cluster.*

The non-compact SSP formulation in `02_noncompact_formulation.tex` looks like a GTSP: the configuration space $\mathcal{C}$ is the vertex set, job clusters H_j play the role of the GTSP clusters, and $d(C_u, C_v) = b - |C_u \cap C_v|$ gives edge costs. The objective is to find a minimum-cost path (cycle through a depot) that visits at least one configuration per job cluster.

However, a direct mapping to a standard GTSP solver is blocked by the **Overlap Problem**: in the SSP, a single configuration C may belong to multiple clusters simultaneously (if $T_i \cup T_j \subseteq C$ for distinct jobs i, j), whereas the GTSP requires *disjoint* clusters. Standard GTSP solvers assume that selecting a vertex satisfies exactly one cluster; applying them naively to overlapping clusters produces incorrect results.

The I-N Transformation: SSP $\to$ GTSP

The Intersecting-to-Nonintersecting (I-N) transformation [Lien et al.(1993)Lien, Ma, and Wah] resolves the Overlap Problem by replicating each configuration once per job it can serve.

Construction

Given a U-SSP instance $\mathcal{I}_{\text{SSP}} = \langle J, T, b, \{T_j\} \rangle$, construct the GTSP instance $\mathcal{I}_{\text{GTSP}} = \langle V^*, \mathcal{E}^*, d^*, \{V_j^*\} \rangle$ as follows.

1. **Vertex replication.** For each job $j \in J$, create a disjoint cluster:
$$V_j^* = \{v_{C,j} \mid C \in H_j\} \quad \forall j \in J.$$
Each configuration C that can process multiple jobs spawns multiple distinct replicas (one per job). This guarantees $V_i^* \cap V_j^* = \emptyset$ for $i \neq j$.
2. **Inter-cluster arc costs.** For two replicas in different clusters, inherit the original metric:
$$d^*(v_{C_u,i},\, v_{C_w,j}) = d(C_u, C_w) = b - |C_u \cap C_w| \quad \forall i \neq j.$$
3. **Zero-cost intra-configuration arcs.** If the same configuration C is replicated in two different clusters V_i^* and V_j^*, connecting the two replicas has zero cost:
$$d^*(v_{C,i},\, v_{C,j}) = b - |C \cap C| = b - b = 0.$$
These zero-cost arcs model the fact that processing multiple consecutive jobs under the *same* magazine configuration incurs no tool switch.

Optimality Preservation

Theorem 2.1 (SSP–GTSP Bijection [Colares and Wagler(2026)])**.** *A minimum-cost Hamiltonian cycle in $\mathcal{I}_{GTSP}$ corresponds to an optimal sequence of configurations for $\mathcal{I}_{SSP}$, with identical switching cost.*

Proof. Any Hamiltonian cycle in $\mathcal{I}_{\text{GTSP}}$ selects exactly one replica $v_{C,j}$ from each cluster V_j^*, which maps to selecting one configuration $C \in H_j$ per job — a valid SSP sequence.

Conversely, in any SSP solution, if the magazine stays at configuration C across a contiguous block of jobs $j_1, j_2, \ldots, j_\ell$, the KTNS policy incurs zero switches. In the GTSP

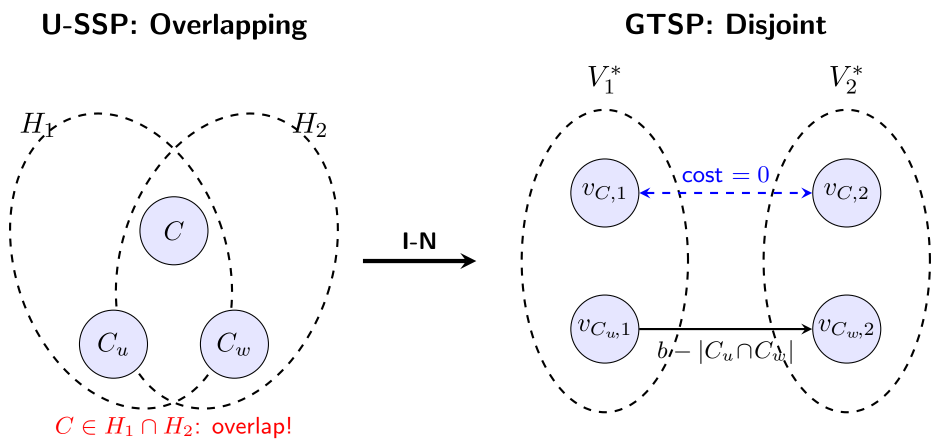

Figure 1: The I-N transformation. Configuration C can process both Job 1 and Job 2 (it is in the intersection $H_1 \cap H_2$). The transformation creates two distinct replicas $v_{C,1} \in V_1^*$ and $v_{C,2} \in V_2^*$, linked by a zero-cost arc. Disjointness is achieved without altering any non-zero transition costs.

graph, the replicas $v_{C,j_1}, v_{C,j_2}, \ldots, v_{C,j_\ell}$ are connected by zero-cost arcs, so the GTSP cycle traverses them at zero cost — exactly mirroring the zero-switch block.

For any transition between distinct configurations C_u and C_w, the arc cost $d^*(v_{C_u,i}, v_{C_w,j}) = b - |C_u \cap C_w|$ equals the number of tool insertions. Thus the GTSP cycle cost equals the total SSP switching cost, and minimising the former minimises the latter. □ □

The Disjointness Condition

The I-N transformation inflates the graph size by replicating configurations. In many instances, however, no replication is needed because the clusters are already disjoint — the SSP is *natively* a GTSP.

Theorem 3.1 (Disjointness Condition [Colares and Wagler(2026)]). *For two distinct jobs $i, k \in J$:*

$$|T_i \cup T_k| > b \implies H_i \cap H_k = \emptyset.$$

Proof. Suppose for contradiction that $H_i \cap H_k \neq \emptyset$. Then there exists $C \in \mathcal{C}$ with $T_i \subseteq C$ and $T_k \subseteq C$, so $(T_i \cup T_k) \subseteq C$. But $|C| = b$, giving $|T_i \cup T_k| \leq |C| = b$, contradicting $|T_i \cup T_k| > b$. □ □

Corollary 3.2 (SSP as a native GTSP). *If $|T_i \cup T_k| > b$ for* every *pair of distinct jobs $i, k \in J$, then $H_i \cap H_k = \emptyset$ for all $i \neq k$, and the U-SSP maps directly to a GTSP with disjoint clusters — no I-N replication is needed.*

Implication. The standard GTSP is a *special case* of the SSP (the disjoint-cluster case), not an independent problem. Whenever instances are "dense" (large $|T_j|$ relative to b), solvers can be applied without the overhead of the I-N transformation.

The Metric Barrier: Limits of the Reverse Reduction

The forward reduction (SSP → GTSP) is polynomial. A natural question is whether an arbitrary GTSP can always be reduced back to an SSP. The answer is no, due to the rigid geometric structure of the SSP metric.

Theorem 4.1 (Metric Barrier [Colares and Wagler(2026)]). *A GTSP instance with arc weight function w can be polynomially reduced to a U-SSP only if w satisfies all four properties simultaneously:*

(i) ***Symmetry:*** *$w(u, v) = w(v, u)$ for all u, v.*
(ii) ***Non-negative integer values:*** *$w(u, v) \in \mathbb{Z}_{\geq 0}$.*
(iii) ***Capacity bound:*** *$w(u, v) \leq b$ for some integer b.*
(iv) ***Triangle inequality:*** *$w(u, v) \leq w(u, x) + w(x, v)$ for all x.*

Proof. Any SSP transition metric $d(C_u, C_v) = b - |C_u \cap C_v|$ inherits these four properties: (i) symmetry from commutativity of $\cap$; (ii) integrality and non-negativity from set cardinalities; (iii) $d \leq b$ since $|C_u \cap C_v| \geq 0$; (iv) the triangle inequality, proved in Part II (`02_noncompact_formulation.tex`, § The Transition Metric).

A GTSP instance violating any one property — e.g., an asymmetric weight $w(u, v) \neq w(v, u)$, a fractional weight, a weight exceeding b, or a triangle inequality violation — cannot be expressed as $b - |C_u \cap C_v|$ for any integer sets C_u, C_v of size b. Since $|C_u \cap C_v|$ is an integer satisfying $0 \leq |C_u \cap C_v| \leq b$, no mapping from such a GTSP to a valid SSP exists. □ □

Remark 4.2. This barrier is not merely theoretical. Real GTSP instances from routing applications often have asymmetric costs (direction-dependent travel times), ruling out direct SSP equivalence. Conversely, SSP instances arising from FMS scheduling always satisfy all four properties by construction.

Dummy Tool Padding and Metric Symmetry

For the restricted GTSP-to-SSP mapping to hold, all configurations must be maximal (size exactly b). Under-capacity states break the symmetry $d(C_u, C_v) = d(C_v, C_u)$ when $|C_u| \neq |C_v|$. Lemma 1 of `01_foundations.tex` ensures this is not an issue: any under-capacity state can be padded to size b with "dummy" tools without increasing cost. Padding with tools from the *next* group's mandatory set (lookahead padding) additionally reduces future transition costs, as formalised in `05_jgp_ssp_gap_analysis.tex`.

Summary of the SSP–GTSP Relationship

Direction	When	Method
SSP $\to$ GTSP (overlapping)	Always	I-N transformation + zero-cost arcs
SSP $\to$ GTSP (disjoint)	$\lvert T_i \cup T_k \rvert > b$ for all $i \neq k$	Direct (no replication)
GTSP $\to$ SSP	Only if metric satisfies (i)–(iv)	Reverse mapping

The I-N transformation is polynomial in $|\mathcal{C}| = \binom{m}{b}$, which is exponential in m and b. In practice, the column generation approach of `02_noncompact_formulation.tex` implicitly exploits the GTSP structure without ever building the full transformed graph.

References

[Colares and Wagler(2026)] R. Colares and A. Wagler. Exact formulations for the job sequencing and tool switching problem. *Working Paper, Université Clermont Auvergne, LIMOS*, 2026.

[Lien et al.(1993)Lien, Ma, and Wah] Y.-N. Lien, E. Ma, and B. W. S. Wah. Transformation of the generalized traveling-salesman problem into the standard traveling-salesman problem. *Information Sciences*, 74(1-2):177–189, 1993.